

BAULKHAM HILLS HIGH SCHOOL

MARKING COVER SHEET



2011

YEAR 12 JUNE ASSESSMENT

EXTENSION 1

STUDENT NAME: _____

TEACHERS NAME: _____

QUESTION	MARK
1	
2	
3	
4	
5	
6	
7	
8	
9	
TOTAL	/
PERCENTAGE	%

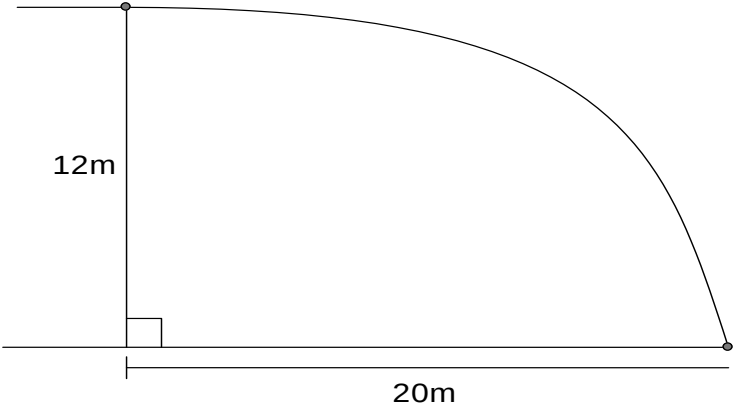
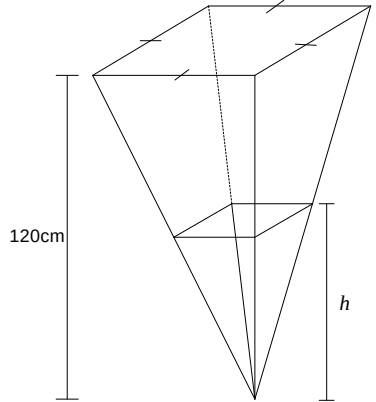
Topics Tested: Further Integration, Newton's Method & Halving the Interval Method,
Applications of Calculus (Related Rates, Velocity as a function of x , Projectile Motion, Simple Harmonic Motion)



YEAR 12 EXTENSION 1 MATHEMATICS ASSESSMENT JUNE 2011

TIME : 60 MINUTES

NAME		RESULT
DIRECTIONS	- Full working should be shown in every question. Marks may be deducted for careless or badly arranged work. - Use black or blue pen only (<i>not pencil</i>) to write your solutions. - No liquid paper is to be used. If a correction is to be made, one line is to be ruled through the incorrect answer.	
QUESTION 1	Evaluate $\int_0^1 6x\sqrt{9-x^2} dx$ using the substitution $u = 9 - x^2$ Answer to 2 significant figures.	4
QUESTION 2	Taking $x = \frac{\pi}{3}$ as the first estimate for the root of $f(x) = 3 \cos x - 4 \tan x$ use one application of Newton's method to find a closer approximation for the root to 2 decimal places.	3
QUESTION 3	A particle is moving along the x axis and its velocity (v) in metres/second at position x metres from the origin is given by $v^2 = 16 + 4x - 2x^2$ (a) Show that the particle undergoes Simple Harmonic Motion. (b) Find the amplitude of the motion. (c) What is the frequency of the motion? (d) What is the maximum acceleration of the particle?	2 2 1 1
QUESTION 4	Find $\int \frac{e^{3x}}{1 + e^{6x}} dx$ using the substitution $u = e^{3x}$	3
QUESTION 5	The displacement of an object as it moves in Simple Harmonic Motion about the origin is given by, $x = 4\sin(3t + \alpha)$ Initially the particle is 2 cm to the right of the origin. Find (i) α (ii) the maximum speed and when it first reaches this speed .	1 4

QUESTION 6	The acceleration of a particle is given by $a = x - 1$, where x is the displacement from the origin. Initially it's at the origin moving with a velocity(v) of 1 m/s (a) Show that $v^2 = (x - 1)^2$ (b) Find the speed of the particle when $x = \frac{1}{2}$ (c) Show that $x = 1 - e^{-t}$ (d) Will the particle reach $x = 2$? Justify your answer.	2 1 3 1
QUESTION 7	A ball is projected horizontally from the top of a 12 metre building. It hits the ground at a point 20 metres from the base of the building  (a) Taking the base of the building as the origin derive the equations of motion Take $g = 10 \text{ m/s}^2$ (b) Find the time that it takes for the ball to hit the ground. (Answer to 1 decimal place). (c) Find the initial speed of the ball. (d) Find the velocity and angle with which the ball hits the ground.	2 1 1 3
QUESTION 8	 Water is poured into a container at a rate of $400 \text{ cm}^3/\text{min}$. The container is an inverted square pyramid of height 120 cm and base edge 60 cm (a) Show that the volume of the water in the container when the depth of the water is $h \text{ cm}$ is given by $V = \frac{h^3}{12}$. (b) Find the rate at which the water level is rising when the depth of the water is 40 cm. (c) Find the time taken for the water level to reach a depth of 100 cm.	2 2 3
QUESTION 9	Find $\int (2\sin x + \cos x)^2 dx$	4
~ END OF EXAM ~		

$$1. \int_0^1 6x\sqrt{9-x^2} dx \quad u = 9-x^2$$

$$\frac{du}{dx} = -2x \rightarrow dx = \frac{du}{-2x}$$

$$\text{when } x=1 \quad u=8$$

$$x=0 \quad u=9$$

$$6x\sqrt{9-x^2} dx = -3 \int_9^8 \sqrt{u} \cdot \frac{du}{-2x}$$

$$\therefore \int_0^1 6x\sqrt{9-x^2} dx = -3 \int_9^8 u^{\frac{1}{2}} \frac{du}{-2x}$$

$$= -\frac{3}{2} \left[\frac{2}{3} u^{\frac{3}{2}} \right]_9^8$$

$$= -2 \left[\sqrt{8^3} - \sqrt{9^3} \right]$$

$$= 8.7$$

$$2. f(x) = 3\cos x - 4\tan x$$

$$3. f'(x) = -3\sin x - 4\sec^2 x$$

$$f\left(\frac{\pi}{3}\right) = 3\cos\left(\frac{\pi}{3}\right) - 4\left(\tan\frac{\pi}{3}\right)$$

$$= -5.428$$

$$f'\left(\frac{\pi}{3}\right) = -3\sin\left(\frac{\pi}{3}\right) - 4\sec^2\left(\frac{\pi}{3}\right)$$

$$= -18.598$$

$$\therefore a_1 = \frac{\pi}{3} - \left(\frac{-5.428}{-18.598} \right)$$

$$= 0.76$$

$$3. a) v^2 = 16 + 4x - 2x^2$$

$$6. \frac{1}{2} v^2 = 8 + 2x - x^2$$

$$a = 2 - 2x$$

$$= -2(x-1)$$

which is in the form ①

$$\ddot{x} = -\omega^2(x-b) \therefore \text{is SHM.}$$

$$b) v=0 \quad -2(x-2x-8)=0$$

$$-2(x-4)(x+2)=0$$

$$x=4, -2$$

$$c) n = \sqrt{2} \therefore \text{freq } \gamma = \frac{n}{2\pi} = \frac{\sqrt{2}}{2\pi} \left(\frac{1}{\pi\sqrt{2}} \right)$$

$$d) \text{Max acc'n when } x = -2$$

$$\ddot{x} = -2(-2-1)$$

$$= 6 \text{ m/s}^2$$

$$4. \int \frac{e^{3x}}{1+e^{6x}} dx \quad u = e^{3x}$$

$$(u^2 = e^{6x})$$

$$\frac{du}{dx} = 3e^{3x} \rightarrow dx = \frac{du}{3e^{3x}}$$

$$\frac{e^{3x}}{1+e^{6x}} dx = \frac{e^{3x}}{1+u^2} \cdot \frac{du}{3e^{3x}}$$

$$\therefore \int \frac{e^{3x}}{1+e^{6x}} dx = \frac{1}{3} \int \frac{1}{1+u^2} du$$

$$= \frac{1}{3} \tan^{-1}(u) + C$$

$$= \frac{1}{3} \tan^{-1}(e^{3x}) + C$$

$$5. (i) x = 4 \sin(3t + \alpha)$$

$$5. \text{when } t=0 \quad x=2$$

$$\text{sub in } \frac{2}{4} = \sin(0 + \alpha)$$

$$\sin \alpha = \frac{1}{2} \rightarrow \alpha = \frac{\pi}{6}$$

$$\therefore x = 4 \sin\left(3t + \frac{\pi}{6}\right)$$

(ii) Centre of the motion is zero

max speed at $x=0$.

$$\text{now } a=4 \quad n=3 \quad x=0$$

$$\text{now } v^2 = 3^2(4^2 - 0^2)$$

$$v = \pm 12$$

$\therefore$ Max speed is 12 cm/s

$$\text{Max Speed when } x=0 \quad 4\sin\left(3t + \frac{\pi}{6}\right) = 0$$

$$3t + \frac{\pi}{6} = 0, \pi, 2\pi, \dots$$

$$3t = \frac{5\pi}{6}$$

$$t = \frac{5\pi}{18}$$

$$t = -\frac{\pi}{18}$$

with working

$$7. \frac{1}{2} v^2 = \int x-1 dx$$

$$= \frac{x^2}{2} - x + C$$

$$\text{when } x=0 \quad v=1$$

$$\therefore \frac{1}{2} = C$$

$$\therefore \frac{1}{2} v^2 = \frac{x^2}{2} - x + \frac{1}{2}$$

$$v^2 = x^2 - 2x + 1$$

$$= (x-1)^2$$

$$b) \text{when } x = \frac{1}{2} \quad v^2 = \left(\frac{1}{2} - 1\right)^2$$

$$v^2 = \frac{1}{4}$$

$$v = \pm \frac{1}{2}$$

$$\therefore \text{Speed} = \frac{1}{2} \text{ m/s}$$

$$c) v^2 = (x-1)^2$$

$$v = \pm(x-1) \text{ but when } x=0 \quad v=1$$

$$\therefore v = -x+1$$

$$\frac{dx}{dt} = 1-x$$

$$\frac{dt}{dx} = \int \frac{1}{1-x} dx$$

$$t = -\ln(1-x) + C$$

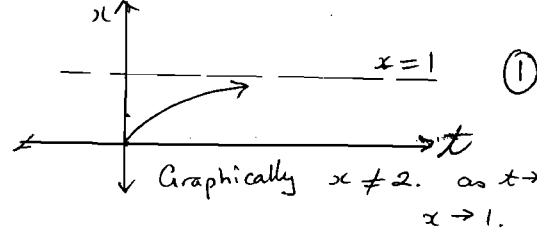
$$\text{when } t=0 \quad x=0 \Rightarrow C=0$$

$$\therefore -t = \ln(1-x)$$

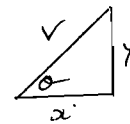
$$e^{-t} = 1-x$$

$$x = 1 - e^{-t}$$

$$d) \text{The graph of } x = 1 - e^{-t}$$



$$7. x=0 \quad y = -10t + C_2$$



$$\begin{aligned} x &= V \cos \theta \\ y &= V \sin \theta \end{aligned} \quad \text{when } \theta = 0$$

$$\begin{aligned} x &= V \\ y &= 0 \end{aligned}$$

$$\therefore \dot{x} = V \quad \dot{y} = -10t$$

$$x = Vt + C_3 \quad y = -5t^2 + C_4$$

$$\text{when } t=0 \quad x=0 \quad y=12$$

$$\therefore x = Vt \quad y = -5t^2 + 12$$

$$\therefore \text{Eqns. } \begin{cases} \dot{x} = 0 \\ \dot{x} = V \\ x = Vt \end{cases} \quad \begin{cases} \ddot{y} = -10 \\ \dot{y} = -10t \\ y = -5t^2 + 12 \end{cases}$$

$$b) \text{when } y=0 \quad -5t^2 = -12$$

$$t = \sqrt{\frac{12}{5}} \quad (t > 0)$$

$$= 1.59 \dots$$

$$\text{①} = 1.6 \text{ s (approx)}$$

$$c) x=20 \text{ when } y=0 \quad t=1.6$$

$$\therefore 20 = V(1.6)$$

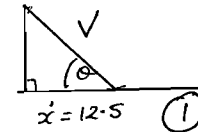
$$V = \frac{20}{1.6} \rightarrow V = 12.5 \text{ m/s}$$

$$(V = 12.9 \text{ m/s})$$

d) At impact

$$\text{at } t=1.6$$

$$\dot{y} = -16 \text{ m/s}$$



$$V = \sqrt{(-16)^2 + (12.5)^2}$$

$$= 20.3 \text{ m/s}$$

$$\tan \theta = \frac{-16}{12.5}$$

$$\text{Angle} = 180 - 52^\circ$$

$$= 118^\circ \quad (\text{Accept acute})$$

$$\text{angle } 52^\circ$$

8. a) 7

$x = \text{square edge}$

$\frac{h}{120} = \frac{x}{30}$ ①

$\therefore x = \frac{h}{2}$

$$V = \frac{1}{3} \times x^2 \times h$$

$$= \frac{1}{3} \times \left(\frac{h}{2}\right)^2 \times h$$

$$V = \frac{h^3}{12}$$

b) Find $\frac{dh}{dt}$ when $h = 40$ cm.
given $\frac{dV}{dt} = 400$

$$\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$$

$$\frac{dV}{dh} = \frac{3h^2}{12} = \frac{h^2}{4}$$

when $h = 40$ $\frac{dV}{dh} = 400$ ①

$$\therefore 400 = 400 \times \frac{dh}{dt}$$

$$\therefore \frac{dh}{dt} = 1 \text{ cm/sec.} \quad \text{①}$$

c) $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$

$$400 = \frac{h^2}{4} \cdot \frac{dh}{dt} \quad \text{①}$$

$$\frac{dh}{dt} = \frac{1600}{h^2}$$

$$\frac{dt}{dh} = \frac{h^2}{1600}$$

$$t = \int \frac{h^2}{1600} dh \quad \text{①}$$

$$t = \frac{h^3}{4800} + c$$

when $t = 0$ $h = 0 \rightarrow c = 0$

$$\therefore t = \frac{h^3}{4800} \text{ when } h = 100$$

① $t = \frac{(100)^3}{4800} \rightarrow t = 208.3 \text{ min}$
(= 3 hrs 28 min)

9. 4 $\int (2\sin x + \cos x)^2 dx$

$$= \int 4\sin^2 x + 4\sin x \cos x + \cos^2 x$$

$$= \int (4\sin^2 x + 2\sin 2x + 1 - \sin^2 x) dx$$

$$= \int 3\sin^2 x + 2\sin 2x + 1$$

$$= \int 3\left(\frac{1}{2} - \frac{1}{2}\cos 2x\right) + 2\sin 2x + 1$$

$$\text{①} \leftarrow \int \frac{5}{2} - \frac{3\cos 2x}{2} + 2\sin 2x$$

$$= \frac{5x}{2} - \frac{3\sin 2x}{4} - \frac{2\cos 2x}{2} + c$$

$$= \frac{5x}{2} - \frac{3\sin 2x}{4} - \cos 2x + c \quad \text{①}$$